

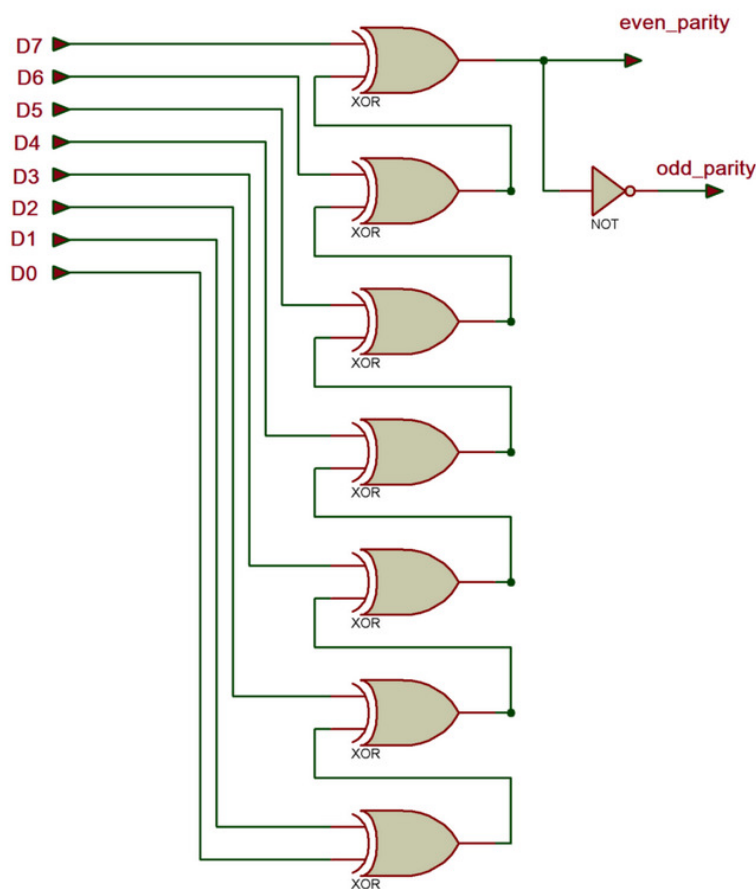
Circuit Simulation Lab:31

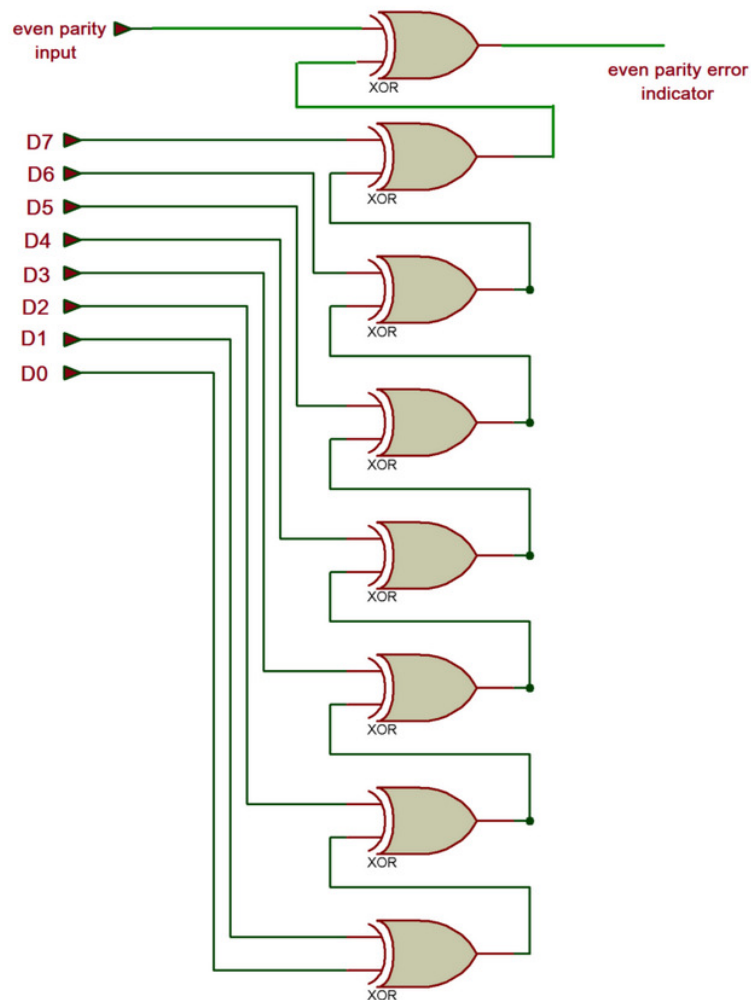
Parity Generator and Checker

Introduction

The purpose of this experiment is to design a Parity Generator and Checker. A Parity Generator is a combinational logic circuit that generates the parity bit in the transmitter. On the other hand, a circuit that checks the parity in the receiver is called Parity Checker. A combined circuit or device of parity generators and parity checkers are commonly used in digital systems to detect the single bit errors in the transmitted data.

8-bit parity generator circuit





8-bit parity Checker circuit

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado

Design 8 bit Parity Generator and Checker Circuit in Verilog HDL and write its testbench and capture the Waveforms.

Use Behavior modeling.